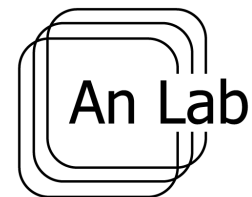


Modeling heterogeneity in single-cell perturbation states enhances detection of response eQTLs

Valencia et al., *Nature Genetics* (2025)

2025.11.12 Journal Meeting

변가은



KOREA
UNIVERSITY

[Kristen brennand] Stem Cell Models for Psychiatric Genetics

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Our work integrates the fields of genomics, neurobiology, and stem cells, defining the disrupted molecular mechanisms that contribute to brain disease, in order to better facilitate the clinical translation of genetic findings. Our focus lies in

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- Psychiatric disease는 유전 + 환경 + Cell state가 복합적으로 작용한다

Psychiatric disease ~ Environment (stress)

- Stress가 psychiatric disease를 증가시키는 것은 하나 단독으로 casual한 것은 아님
- ASD, Schizophrenia, BD 등은 heritability ~70% 이상, MDD나 PTSD는 30% 정도이며, 환경적인 요소 (stress)도 발병 원인의 큰 부분을 차지함
 - Schizophrenia → rare and common variants / PTSD → common variants / other psychiatric disorder → highly penetrant coding mutation 등 Disease마다 genotype 종류에 따른 영향 다르고, 그 안에서도 patient 마다 genetic risk도 모두 상이함.
 - Environmental 요소가 이러한 genetic phenotype 발현에 영향을 줄 것이다.
 - 그러면 Does Environment make phenotype worse, more expressive and more penetrant?
- PTSD 환자와 control을 비교하여 neuron에서 특이적인 stress gene 발굴
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 - Stress가 neuron에 주는 영향이 psychiatric disease 전반에 걸쳐 dysregulation 역할을 함

Environmental Triggers

- How do stress impact on gene expression?
- iPSC derived neuron에 염증/스트레스 자극 (e.g IL-6 / IFN-α / Hcort)
 - *2nd trimester에 이러한 자극이 있으면 특히 더 위험할 수 있음
- MIA 모델링
 - 단순 '스트레스 여부'가 아니라 자극 종류별로 유전자 반응 다름
 - 특정 SNP가 자극 상황에 따라 반응 양상이 달라짐 (stress-interacting eQTL)
 - PTSD iPSC 뉴런 저농도 스트레스에도 과민반응
 - environ.은 risk gene 발현을 유도하는 동적 상태로 작용한다
 - 염증 자극에 따라 조절되는 유전자들 중 상당수가 정신질환 GWAS와 겹침
 - 세포 상태에 따라 유전자 반응성과 위험도가 달라짐
 - 실제 사이토카인이 어떻게 태아의 뇌에 전달되는지는 아직 모른다
 - Placenta, BBB 고려하면 in vitro 모델로는 전체 과정 재현 어렵다

유전 - 환경 상호작용

- 특정 genetic variant가 stress환경에서만 유전자 발현에 영향을 줄 수 있나?
 - PTSD 모델링 (iPSC)
- Stress-interacting eQTL (iPSC + postmortem brain)
 - TF (YY1, NF-κB 등) 조절 실험함
 - 동일 SNP도 스트레스 여부에 따라 유전자 조절 방향 달라짐
- PTSD 환자 뇌 조직 vs. 코르티솔 처리 iPSC 뉴런 반응의 50% 이상 일치
 - 예상보다 더 비슷해서 놀랐다고 한다
 - YY1 knockdown 시 → 스트레스 반응 유전자 발현 완화됨
- 유전자 간 convergence → 공통 downstream 타겟 분석
 - ex. Estradiol treatment

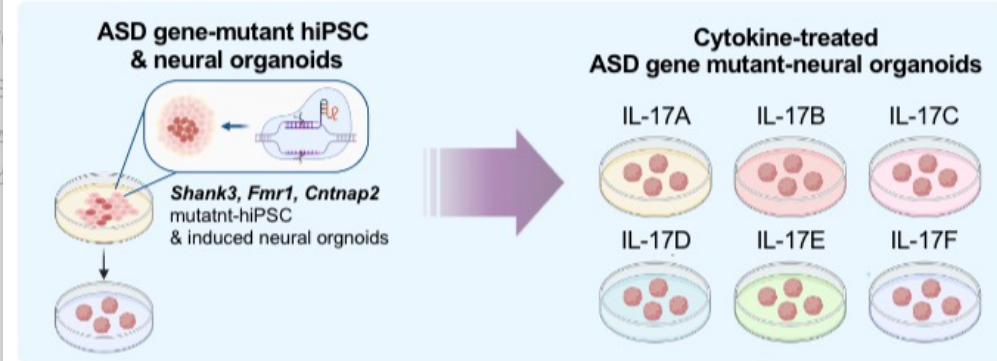
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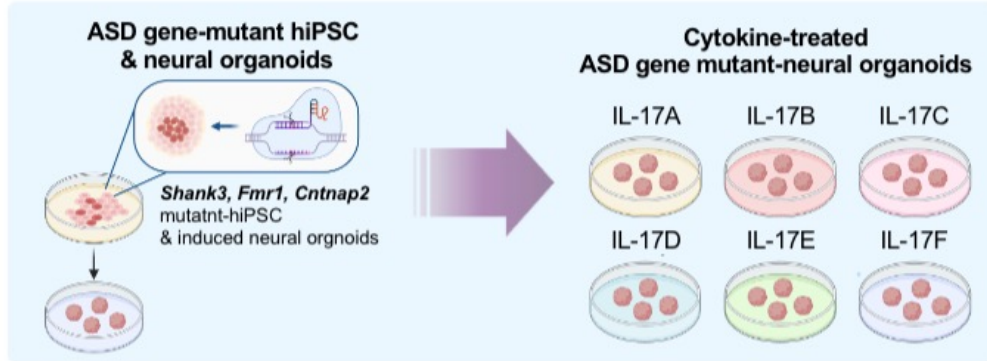
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Environmental Triggers

nature genetics



Analysis

<https://doi.org/10.1038/s41588-025-02344-6>

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[Check for updates](#)

Cristian Valencia^{1,2,3,4,5}, Aparna Nathan^{1,2,3,4,5}, Joyce B. Kang^{1,2,3,4,5}, Laurie Rumker^{1,2,3,4,5}, Hyunsun Lee^{2,5,6,7} & Soumya Raychaudhuri^{1,2,3,4,5}✉

Identifying response expression quantitative trait loci (reQTLs) can help to elucidate mechanisms of disease associations. Typically, such studies model the effect of perturbation as discrete conditions. However, perturbation experiments usually affect perturbed cells heterogeneously. Here we show that modeling of per-cell perturbation state enhances detection of reQTLs.

We use single-cell data to study the effect of perturbations with influenza A virus, *Canidia albicans*, *Pseudomonas aeruginosa* and *Mycobacterium tuberculosis* on gene regulation. We found on average 36.9% more reQTLs by accounting for single-cell heterogeneity compared to the standard discrete reQTL model. For example, we detected a decrease in the expression quantitative trait loci effect for *PXK* with influenza A virus. Furthermore, we found that, on average, 25% of reQTLs have cell-type-specific effects. For example, the reQTL effect for *RPS26* was stronger in B cells. Our work provides a general model for more accurate reQTL identification and underscores the value of modeling cell-level variation.

Genome-wide association studies (GWAS) have elucidated loci

One way to understand context-dependant gene regulation is

- ex. Estradiol treatment

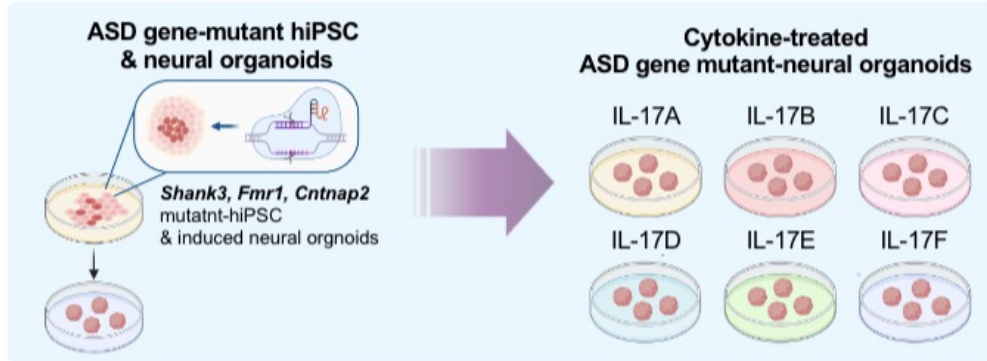
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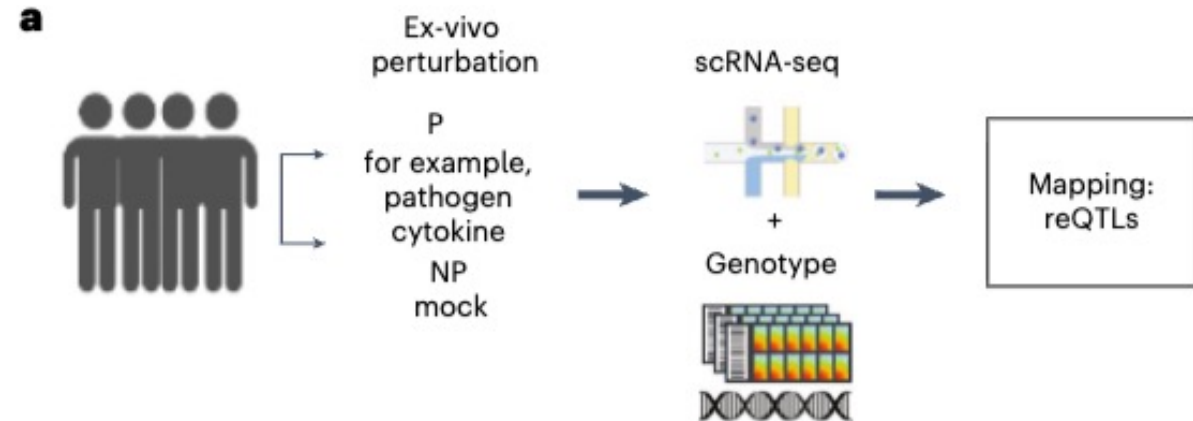
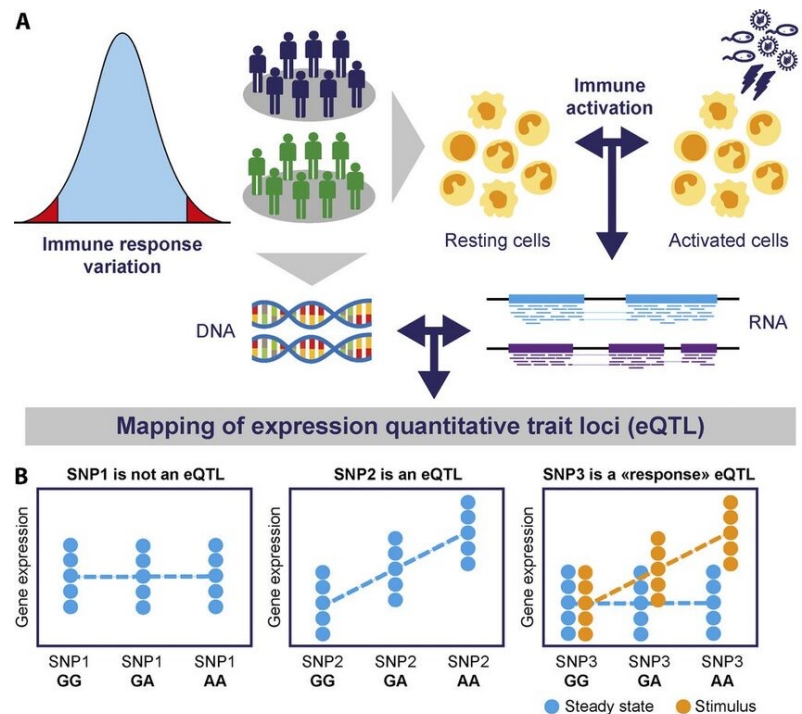
- ex. Estradiol treatment

Introduction

- Less than 10% of eQTLs found at homeostasis colocalize with GWAS loci, suggesting that disease-associated regulation **may emerge only in specific cellular contexts** such as those critical for disease initiation.
- Previous reQTL models typically only offer a **single format** to represent perturbation response.

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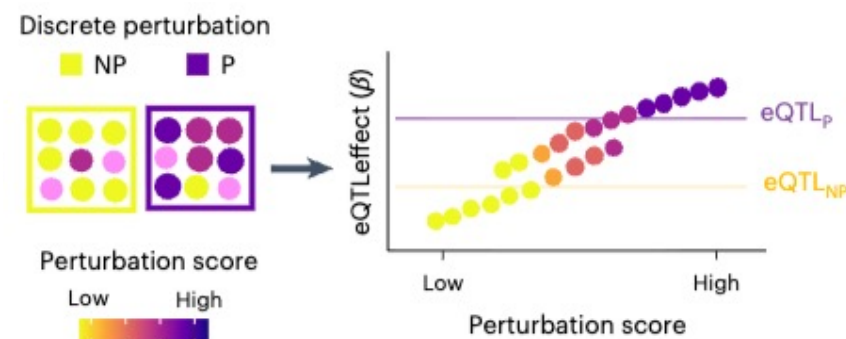
H1: $\beta_G \times \text{Discrete} \neq 0$

Introduction

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- Previous reQTL models typically only offer **a single format** to represent perturbation response.
- We hypothesize that effective modeling of **per-cell perturbation state using linear models** may enhance power to detect response eQTLs.

💡 Continuous perturbation states (2df-model)

	기존 reQTL 방식 (1df-discrete model)	Valencia 2025 (2df-model)
Perturbation state 정의 방식	Discrete	Discrete + Continuous
	세포가 교란됨 (P=1) 또는 교란되지 않음 (NP=0)의 두 가지 조건으로 분류됨	(1) 이산적 상태 (GxDiscrete)+ (2) 연속 교란 점수 (GxScore) 두 가지 상호작용 항을 동시에 포함하여 모델링
Test	외부 교란 (P vs NP) 후 유전자 발현에 미치는 유전적 효과(eQTL)의 크기 변화를 확인	Perturbation 의 이질적인 정도(연속적인 반응)에 따른 유전적 효과 포착.



Statistical model for response eQTL detection

eQTL

$$\log(E_i) = \theta + \beta_G X_{d, \text{geno}} + \beta_{\text{age}} X_{d, \text{age}} + \beta_{\text{nUMI}} X_{i, \text{nUMI}} + \beta_{\text{MT}} X_{i, \text{MT}}$$

reQTL

$$+ \beta_{\text{Discrete}} X_{i, \text{Discrete}} + \beta_{\text{rScore}} X_{i, \text{rScore}} + \beta_{\text{GxrScore}} X_{d, \text{geno}} X_{i, \text{rScore}}$$

$$+ \beta_{\text{GxDiscrete}} X_{d, \text{geno}} X_{i, \text{Discrete}} + \sum_{k=1}^5 \beta_{\text{gPC}_k} X_{d, \text{gPC}_k} \quad (2)$$

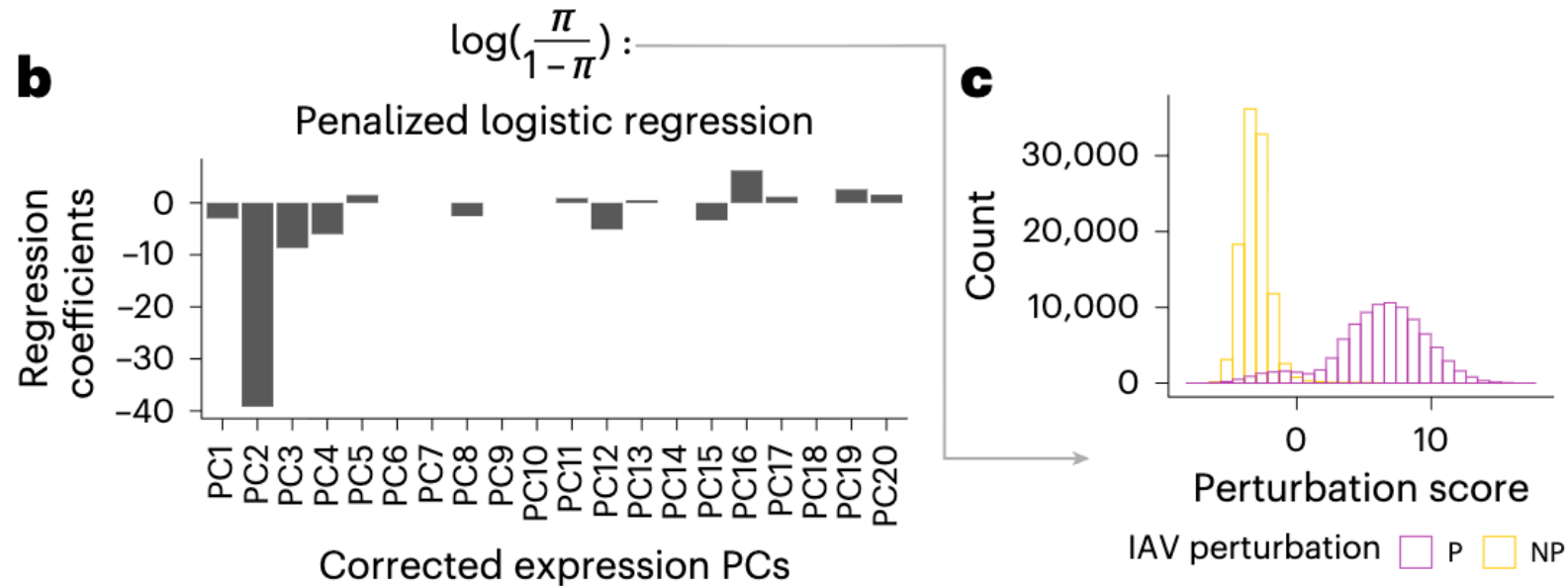
$$+ \sum_{k=1}^5 \beta_{\text{ePC}_k} X_{i, \text{ePC}_k} + (d) + (k_b | b),$$

$$\text{Score} = \theta + \beta_{\text{Discrete}} X_{i, \text{Discrete}} + r \text{ Score}. \quad (3)$$

💡 Statistical Framework for Response eQTL Mapping

→ Modeling genotype × perturbation interactions in single cells

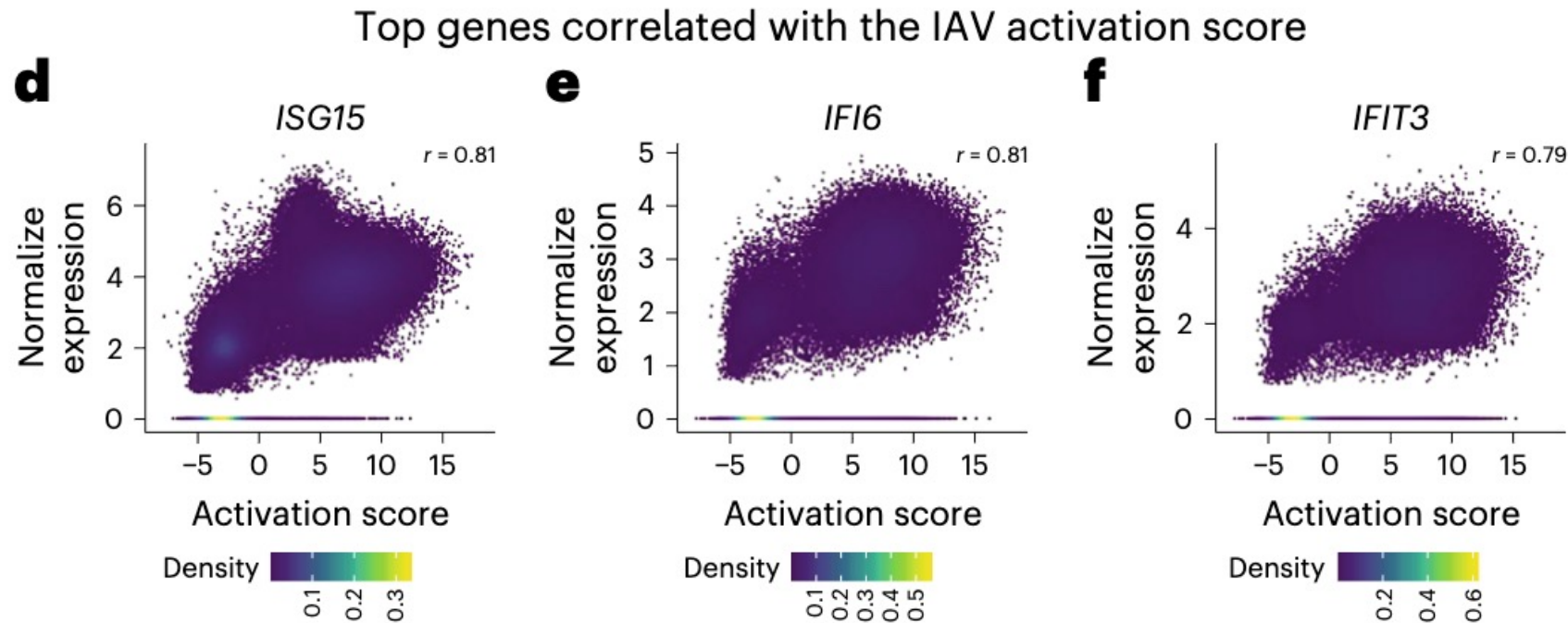
- 💡 Predicting the log odds of being in the pool of perturbed (P) cells hPCS (LASSO regression)
- 💡 Identifying response eQTL effects using a Poisson mixed effects model of GEX in each cell



Result 1

Quantifying perturbation states

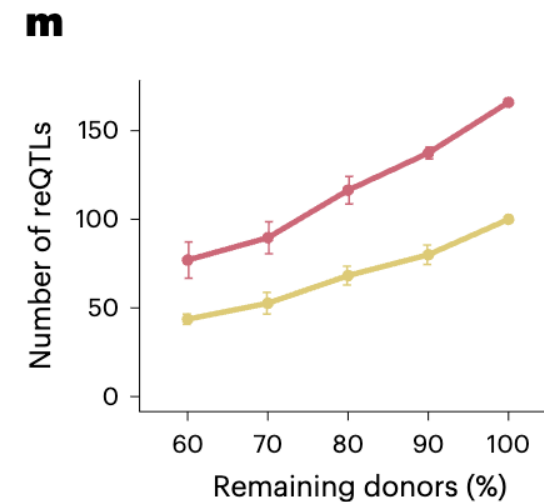
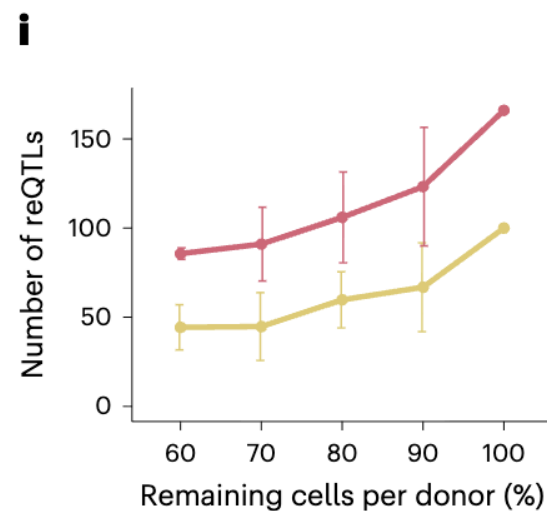
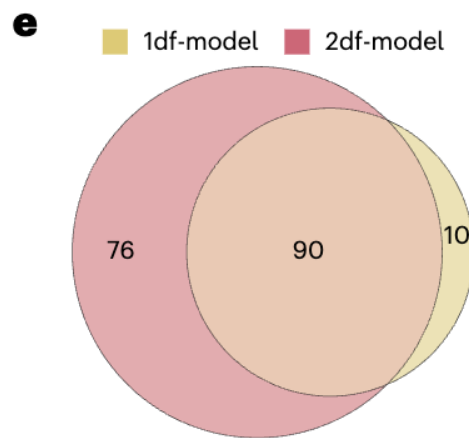
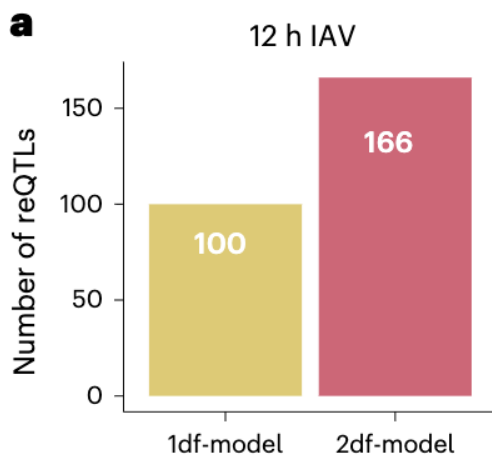
- Perturbation: **Influenza A virus (IAV)**, *Candida albicans* (CA) (PBMC)
- IAV perturbation → ISG15/ IFI6 (IFN- α response related gene) GEX
- hPCs를 활용해 정의하여 계산한 Perturbation score가 transcriptional heterogeneity를 잘 (Pearson corr. = 0.81) 반영함



Result 2

Mapping reQTLs

- 기존 1df-model의 reQTL이 재현되었고 Pertub. 조건 평균 36.9%의 새로운 reQTLs을 발견함
- We observed a linear power loss by reducing donors and cells.



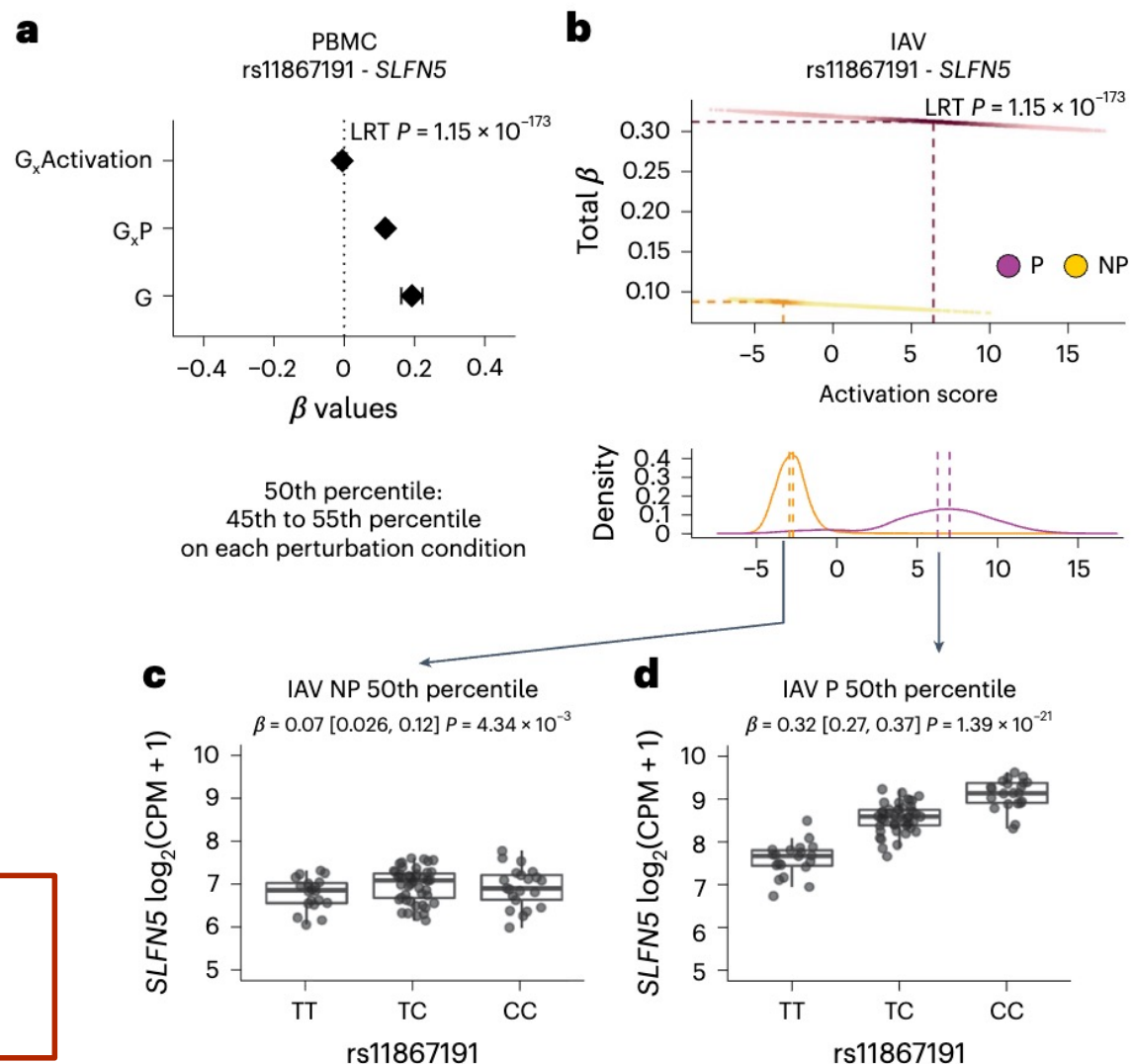
Result 4

reQTLs with strong perturbation effects

- rs11867191** on *SLFN5*

- a) $G \times \text{Activation}$, $G \times P$ interaction term $\rightarrow$ Total effect에 기여
- b) Activation (Perturbation) score $\sim \beta_{\text{total}}$
- c) NP 45~55th percentile 에 해당하는 샘플별 세포 평균 발현 eQTL
- d) P 45~55th percentile 에 해당하는 샘플별 세포 평균 발현 eQTL

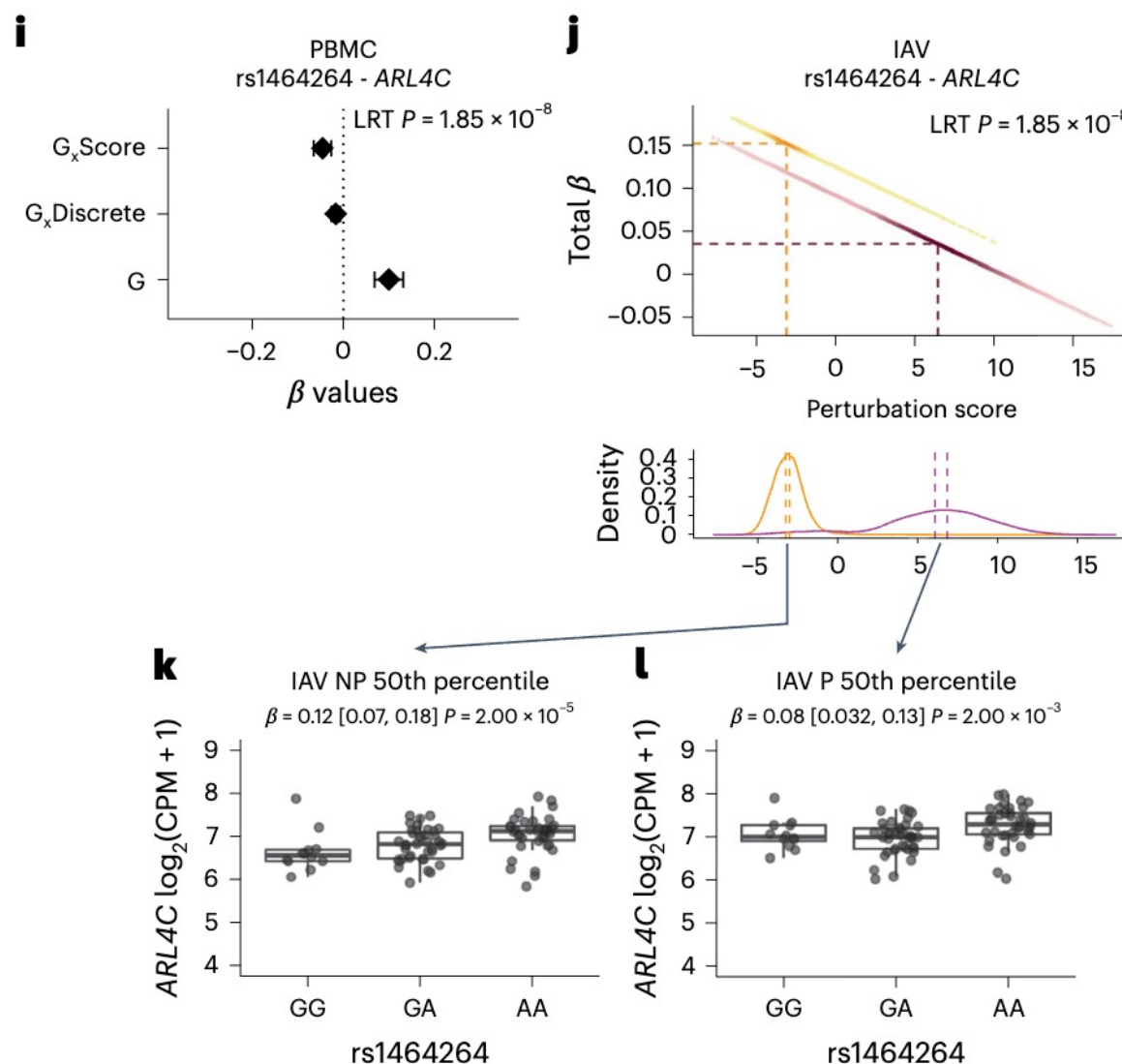
rs11867191 변이는 *SLFN5* 발현에 미치는 영향이
평상시(NP)보다 IAV 자극 후(P) 대폭 증가함
= eQTL의 효과가 자극에 의해 증폭



Result 4

reQTLs with subtle perturbation effects

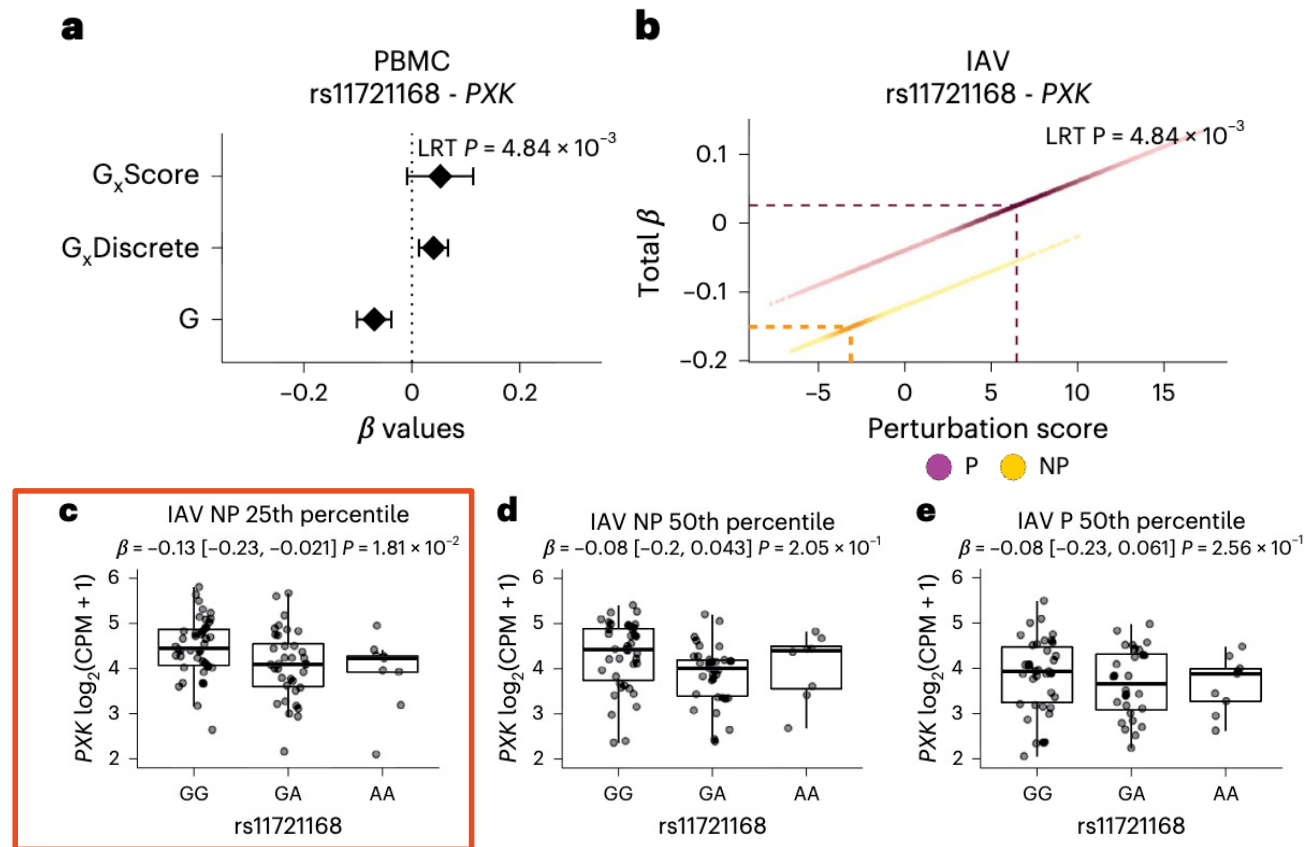
- **rs1464264** on *ARL4C*
 - Was not detected by the PB-linear mixed effect model
 - 1df-discrete 모델로도 변화는 있지만 크지 않음.
 - 2df 모델로 더 명확히 파악.



Result 5

reQTLs can have state-specific colocalization

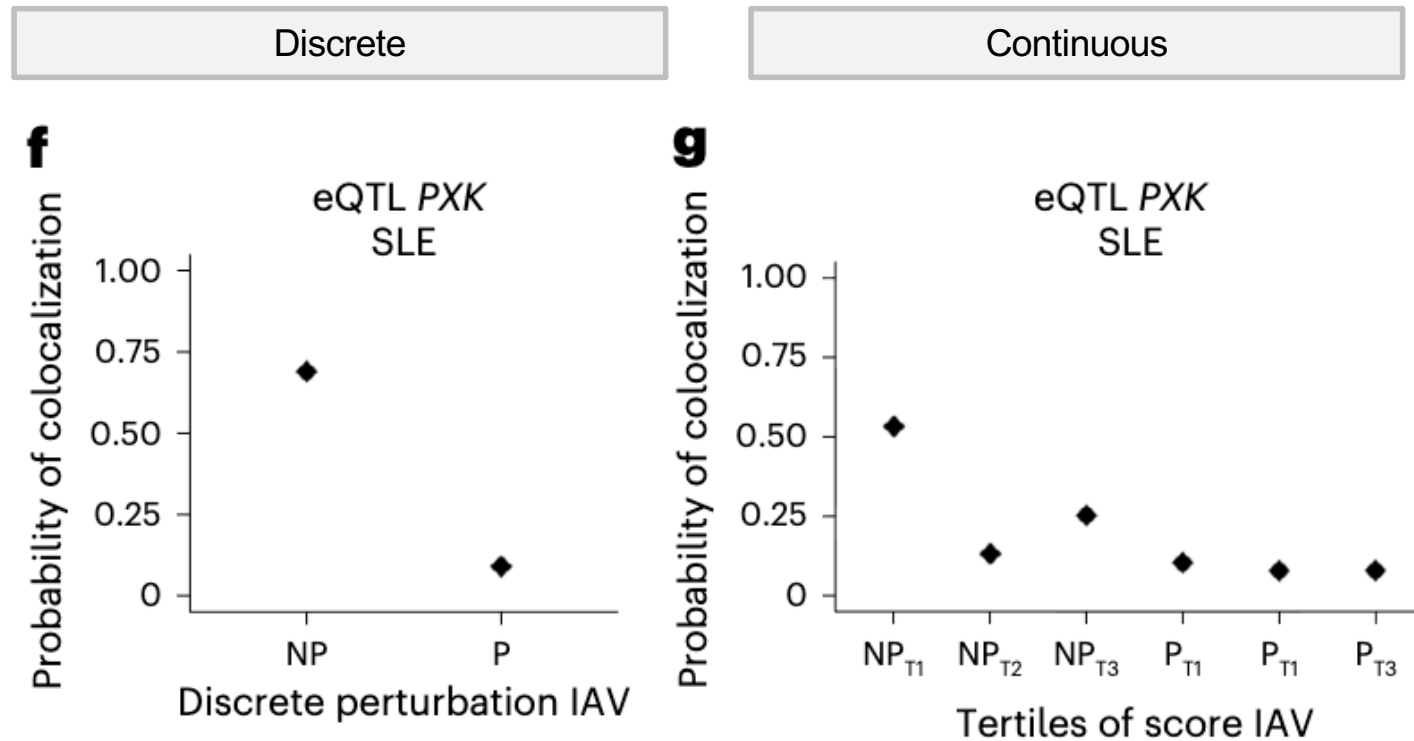
- **rs11721168** on PDK
- Decreased eQTL effects after IAV perturbation
- The effect was detected only in cells that score lower in the NP states



Result 5

reQTLs can have state-specific colocalization

- The eQTL signal for PXX colocalized in the lowest perturb. tertile of the NP cells



Discussion

Summary / Discussion

Summary

- Improved the identification of reQTLs with traits
- 오가노이드 등의 데이터에의 응용

Limitation

- PCA based → Generalization
- Log-linear based → missing non-linearity
- scRNA-seq은 CITE-seq 이 없어서 사용된 것으로 추후 확장 필요